ATOMIC-LEVEL MATERIALS SIMULATION

Module Code: MES306TC

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The Wilson Loop Method and Numerical $\mathbb{Z}_2$ Classification

I. The Physics of Wilson Loops and the Wannier Charge Center (WCC)

II. The $\mathbb{Z}_2$ Criteria and Intersection Counting

III. Numerical Implementation and Troubleshooting

I. The Physics of Wilson Loops and the Wannier Charge Center (WCC)

While the Chern number works for 2D Hall insulators, $\mathbb{Z}_2$ systems (with TRS) require a subtler approach

The Wilson Loop provides the 'Wannier representation' of the bulk bands

A product of overlap matrices between adjacent *k* points

A Wilson Loop is the parallel transport of the occupied states along a closed loop in the reciprocal space

$$W(k_y) = Pe^{i\oint A(k)dk_x}$$

I. The Physics of Wilson Loops and the Wannier Charge Center (WCC)

A $N\times N$ matrix (for NN occupied bands) that encodes the geometric phase acquired by Bloch states as they are transported along a closed loop in momentum space at a fixed k_x. Its eigenvalues give the Wannier center positions (WCCs)

It converts the integrated Berry connection (which is a matrix) into a unitary matrix. In practice, this exponential is never computed directly; it is replaced by a product of overlap matrices in numerical implementations

$$W(k_y) = Pe^{i\oint A(k)dk_y}$$

Path-ordering operator

$A(k)$ is a matrix and its values at different k points do not commute, so the product along the path must be taken in the exact sequential order. This is analogous to time-ordering in quantum mechanics

Closed-loop integral along k_y – the integral runs over the entire Brillouin zone in the k_y direction, returning to the starting point. Because the Brillouin zone is a torus, this is a closed loop for each fixed k_x

$$A_n(k) = i\langle u_n(k)|\nabla_k|u_n(k)\rangle$$

$A_n(k)$ measures how the occupied Bloch states change as k varies. It is gauge-dependent, but its path-ordered exponential (the Wilson loop) is gauge-covariant

I. The Physics of Wilson Loops and the Wannier Charge Center (WCC)

The Wannier Charge Center (WCC)

The eigenvalues of the Wilson Loop matrix are $e^{i\theta_j(k_x)}$

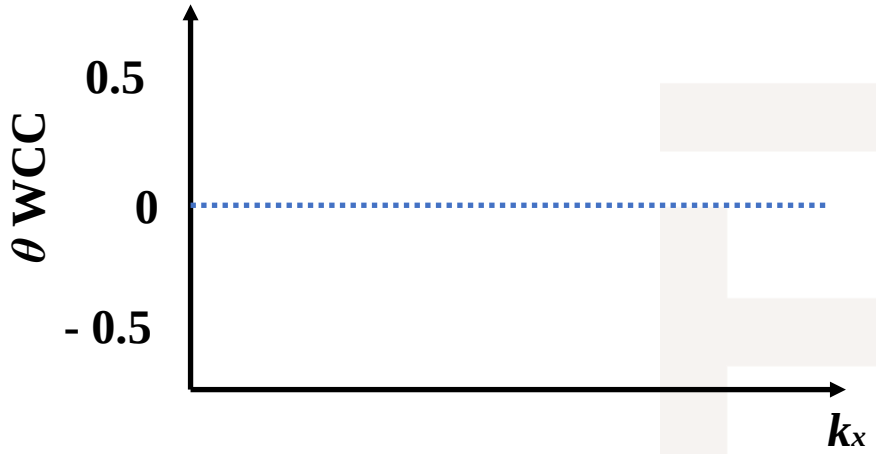
The phases $\theta_j(k_x)$ are the Wannier Charge Centers

They represent the average 'position' of electrons in the Wannier basis

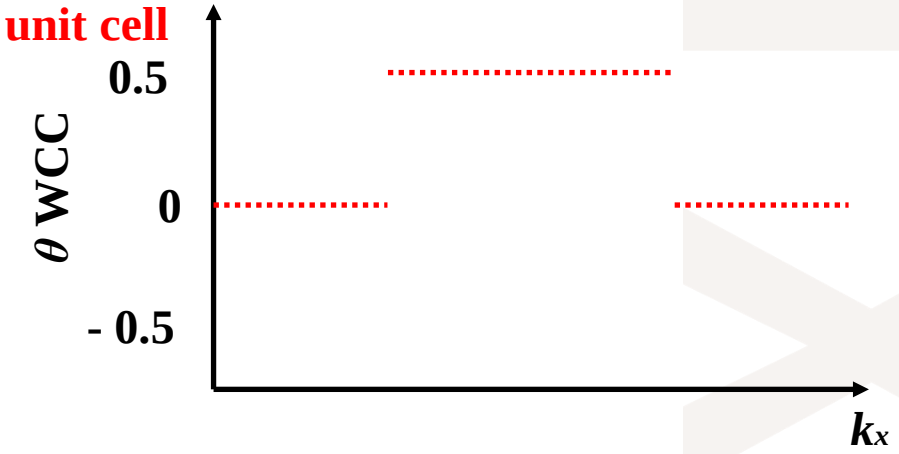
I. The Physics of Wilson Loops and the Wannier Charge Center (WCC)

The WCCs represent where the charge is 'concentrated' in the unit cell

In a trivial insulator, they are localized and flat



In a topological insulator, they are 'pushed' to the boundaries of the unit cell



I. The Physics of Wilson Loops and the Wannier Charge Center (WCC)

Visualizing the WCC Evolution

As we vary k_x, the WCCs trace out paths in the (θ, k_x) plane

This is the Wilson Loop Spectrum

II. The $\mathbb{Z}_2$ Criteria and Intersection Counting

Time-Reversal Invariant Momenta (TRIM)

In a 2D Brillouin Zone, there are 4 points where $k = -k$: (0, 0), (0, π), (π, 0), (π, π)

At these points, Kramers theorem forces the eigenvalues to be degenerate

Every state at a Time-Reversal Invariant Momentum (TRIM) point must be at least twofold degenerate

II. The $\mathbb{Z}_2$ Criteria and Intersection Counting

At $k_y = 0$ and $k_y = \pi$, the WCC spectrum is strictly pinned in pairs

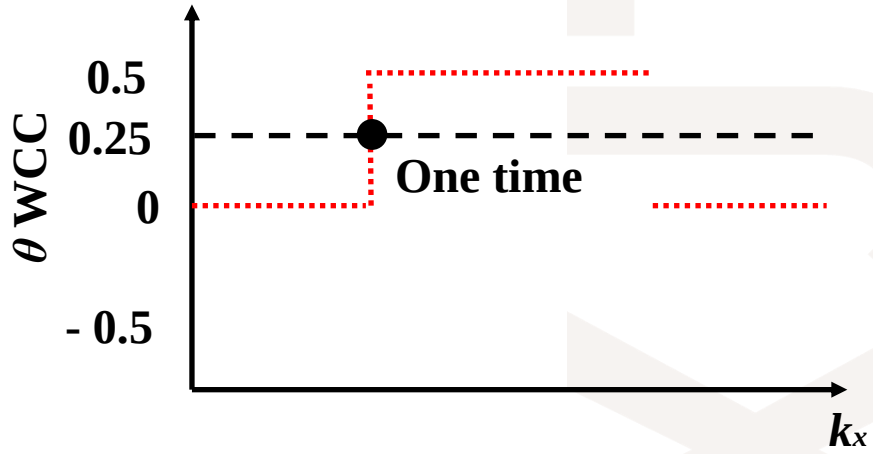
If a curve starts at θ and ends at $\theta+\pi$, it must have crossed the middle of the gap

Choose an arbitrary reference line $\theta = \pi$ (or any value in the gap)

Count how many times the WCC curves cross this line

Non-Trivial — Odd number of crossings (*e.g.* 1, 3, ...): $\mathbb{Z}_2 = 1$

Trivial — Even number of crossings (*e.g.* 0, 2, ...): $\mathbb{Z}_2 = 0$

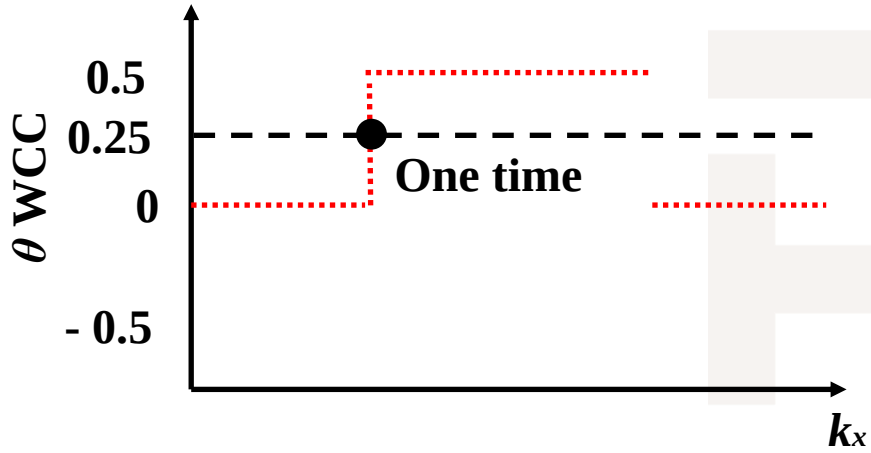


II. The $\mathbb{Z}_2$ Criteria and Intersection Counting

Topological phase transitions are characterized by the 'Partner Switching' of WCC curves

It is a global change in the band connectivity

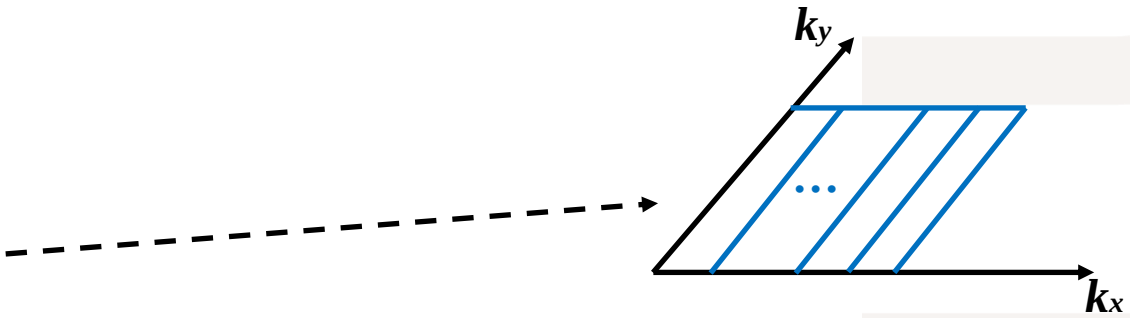
The crossing is protected by TRS



Small perturbations cannot remove the crossing point because it is a topological requirement

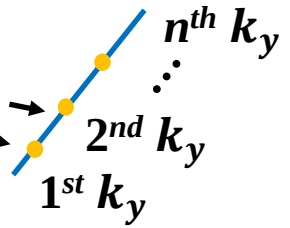
III. Numerical Implementation and Troubleshooting

Computational Procedure:



1. Diagonalize $H(k)$ at each point

2. Compute overlap matrices $M_{mn} = \langle u_m(k_i) | u_n(k_{i+1}) \rangle$



3. Construct the product matrix $D(k_y)$

$$D_{k_y} = \prod M_{i,i+1} \qquad D_{k_y} = \begin{vmatrix} \square & \cdots & \square \\ \square & \cdots & \square \\ \vdots & \vdots & \vdots \\ \square & \cdots & \square \end{vmatrix}$$

4. Eigen-decompose $D(k_y)$ to get phases θ_j

Eigen value $(a+ib)$ $\theta_j = \text{ARCTAN}(b/a)$

III. Numerical Implementation and Troubleshooting

Note:

Numerical codes yield random phases for eigenvectors

The Wilson Loop is gauge-invariant, but the tracking of WCCs is not

To plot a continuous curve, you must match the states between k_y and $k_y + \Delta k_y$

$$M_{mn} = \langle u_m(k_i) | u_n(k_{i+1}) \rangle$$

III. Numerical Implementation and Troubleshooting

Note:

θ is defined in [-π, π]

Prevent 'jagged' vertical lines in the plots

If the bands are very narrow, crossing detection may be ambiguous

Vary the reference line height. If the crossing count changes, the bands might be touching (metal phase)



That's all for this session

Thank you for your attention and great participation